

Kamaljeet Singh

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RESEARCH INTEREST

Since the beginning of my Ph.D., my research has been dedicated to the study of strongly interacting matter within the ALICE experiment at the Large Hadron Collider (LHC) at CERN. I am particularly interested in how matter behaves at extreme temperatures and energy densities, where ordinary hadrons dissolve into a deconfined Quark–Gluon Plasma (QGP). Within this broad framework, my work focuses on the production and polarization of the ϕ meson ($s\bar{s}$) in heavy-ion collisions, analysed through the forward-rapidity dimuon decay channel. The cleanliness of this channel and its weak sensitivity to final-state interactions allow a robust reconstruction of early collision dynamics. Enhanced ϕ meson production in Pb–Pb collisions relative to pp indicates increased strangeness content in the medium, consistent with the formation of a QCD matter. In heavy-ion phenomenology, I am studying the leading and higher order thermoelectric phenomenon in magnetized QCD matter in the context of heavy-ion collisions. I have study the sensitivity of thermoelectric figure of merit to the deconfinement phase transition. Hence, the thermoelectricity is the key window to visualize the intricate dynamics of the QCD matter.

Building on this foundation, my future research plan is to develop a unified and interdisciplinary framework of thermoelectric transport spanning from QCD matter to condensed matter systems. The underlying similarities in transport phenomena, such as charge–heat coupling, collective behavior, and response to external fields, provide a unique opportunity to bridge high-energy nuclear physics with functional materials science. This approach will enable the translation of fundamental insights from QCD thermoelectricity into the design principles of advanced thermoelectric materials thereby connecting microscopic theory with device-level applications.

In parallel, I aim to contribute to the design and development of next-generation beampipe materials through advanced alloy engineering. By leveraging my prior experience in designing ternary alloy systems, I plan to optimize lightweight, high-strength, and thermally stable materials suitable for modern accelerator environments, ensuring improved performance and long-term reliability.

ACADEMIC PROFILE

◦ **Doctor of Philosophy (Ph.D.) Physics** (2021 – present)

- **Thesis Supervisor:** [Prof. Raghunath Sahoo](#)

Department of Physics, Indian Institute of Technology Indore, India

- **Specialization:** Experimental High Energy Physics basically in the context of heavy-ion collisions with ALICE at LHC.
 - **Dissertation:** Study of ϕ meson production in Pb–Pb collision in Run-3 data at 5.36 TeV energy via dimuon decay channel and Study of leading and higher-order thermoelectric transport phenomenon of magnetized QCD matter.
 - **Programming:** I have get expertise in the coding languages C, C++, Python, ROOT during my research.
 - **Event Generators:** I have participate in my research group to simulate the heavy-ion collisions using event generators PYTHIA (for proton-proton collision), A Multiphase Transport Model- AMPT (for Pb-Pb and Au-Au collisions).
- **Masters of Science in Physics** (2016 – 2018) - Guru Nanak National College, Panjab University, Chandigarh, India
 - **Subjects:** Physics, Computational applications of Mathematical Physics.
 - **Programming:** Learnt the advance level of coding language C, C++, Python.
 - **Bachelor of Science** (2012 – 2015) - Sathish Chandra Dhawan Government College, Panjab University, Chandigarh, India
 - **Subjects:** Physics, Chemistry, Math.
 - **Programming:** Basics of coding C, C++.

SCHOLASTIC ACHIEVEMENTS

- **Senior Research Fellowship - Physical Sciences** (2023) - Promoted to Senior Research Fellow after qualifying the Comprehensive exam as per the norms of University Grant Commission. It officially allows me to start working with Junior Research Fellows
- **Junior Research Fellowship - Physical Sciences** (2021) - Qualified for Lectureship and Junior Research Fellowship. The number of appeared candidates for this exam was 30,631. I have cracked this exam and scored all India rank (AIR) - 169. The University Grant Commission funded me for five years to proceed further for my doctorate in experimental high energy physics
- **Graduate Aptitude Test in Engineering (GATE) - Physics** (2021)
- **Graduate Aptitude Test in Engineering (GATE) - Physics** (2020). The number of appeared candidates for this exam was 16,990. I achieved score of 397.
- **Joint Entrance Screening Test (JEST) - Physics** (2020). The number of appeared candidates for this exam was 14,000. I have scored all India rank (AIR) - 305.
- **Bhabha Atomic Research Center (BARC) Scientific officer Exam** (2020) - I have cracked a written exam for the post of an scientific officer conducted by BARC- department of science and technology (DST), Gov. of India.

RESEARCH EXPERIENCE

- **Physics Analysis:** Production analysis of the ϕ meson in Pb–Pb collisions at $\sqrt{s_{\text{NN}}} = 5.36$ TeV using Run–3 ALICE data in the forward-rapidity dimuon channel. The dimuon mode provides a clean decay signature with minimal final-state interactions, allowing a robust measurement of in-medium vector meson modifications.

Along with this analysis, I am also contributing to the study of ϕ -meson production and polarization in pp collisions at $\sqrt{s} = 13.6$ TeV, focusing on signal extraction, angular decay analysis, and cross-section systematics in the forward-rapidity dimuon channel.

- **O² and Hyperloop Workflow:** Active use of ALICE’s O² analysis framework for Run–3 data processing, including asynchronous reconstruction outputs, distributed Hyperloop workflows, and large-scale train-based production for signal extraction and validation.
- **Service Task – MUON and GLOBAL-MUON QC:** Working with [Andrea Ferrero](#), I have Contribute to Run–3 asynchronous Quality Control for both the MUON (MID–MCH) and the Global-Muon (MID–MCH–MFT combined) workflow. Tasks include detector-performance monitoring, occupancy and efficiency checks, trigger-stability validation, run-by-run flagging of problematic fills, and identification of hardware/software anomalies within the O² online–offline system.
- **Phenomenology:** Alongside my experimental work, I have actively contributed to phenomenological studies on transport and thermoelectric properties of QCD matter. This includes calculations of electrical and thermal conductivity in hadronic and quark–gluon matter, analyses of induced electromagnetic fields in thermoelectric QGP, leading and higher-order thermoelectric effects, the thermoelectric figure of merit across the QCD phase transition, and coupled transport phenomena. These studies provide qualitative insights into heat-charge transport and early-time field dynamics. I have also pursued interdisciplinary research on beam-pipe material development for low-energy accelerators, with ongoing work relevant to high-energy facilities.

ANALYSIS NOTES

- ϕ meson production in Pb–Pb collisions at $\sqrt{s_{\text{NN}}} = 5.36$ TeV in forward rapidity with ALICE at LHC.

Kamaljeet Singh, Kangkan Goswami and Raghunath Sahoo

<https://alice-notes.web.cern.ch/node/1690>.

- Production and spin alignment of ϕ meson in pp collision at 13.6 TeV in dimuon channel.

Kangkan Goswami, Dushmanta Sahu, **Kamaljeet Singh**, and Raghunath Sahoo

<https://alice-notes.web.cern.ch/node/1692>.

PUBLICATIONS

- Effect of time-varying electromagnetic field on Wiedemann-Franz law in a hot hadronic matter

Kamaljeet Singh (Indian Inst. Tech., Indore), Jayanta Dey (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore), Sabyasachi Ghosh (Indian Inst. Tech., Bhubaneswar)

DOI: 10.1103/PhysRevD.108.094007 (publication)

Published in: Phys.Rev.D 108, 094007 (2023)

Impact factor: 5.3

- Thermal conductivity of evolving quark-gluon plasma in the presence of a time-varying magnetic field

Kamaljeet Singh (Indian Inst. Tech., Indore), Jayanta Dey (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

DOI: 10.1103/PhysRevD.109.014018 (publication)

Published in: Phys.Rev.D 109, 014018 (2024)

Impact factor: 5.3

- Electric field induction in quark-gluon plasma due to thermoelectric effects

Kamaljeet Singh (Indian Inst. Tech., Indore), Jayanta Dey (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

DOI: 10.1103/PhysRevD.110.114051 (publication)

Published in: Phys.Rev.D 110, 114051 (2024)

Impact factor: 5.3

- Design and development of an advanced material for beam pipe applications in particle accelerators

Kamaljeet Singh (Indian Inst. Tech., Indore), Kangkan Goswami (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore), Sumanta Samal (Indian Inst. Tech., Indore)

DOI: 10.1103/PhysRevAccelBeams.28.043101 (publication)

Published in: Phys. Rev. Accel. Beams 28, 043101 (2025)

Impact factor: 1.8

- Thermoelectric effects of an interacting hadron gas in the presence of an external magnetic field

Kamaljeet Singh (Indian Inst. Tech., Indore), Kshitish Kumar Pradhan (Indian Inst. Tech., Indore), Dushmanta Sahu (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

DOI: 10.1103/PhysRevD.111.074033 (publication)

Published in: Phys. Rev. D 111, 074033 (2025)

Impact factor: 5.3

- Thermoelectric Thomson coefficient of quark-gluon plasma in the presence of a time-varying magnetic field

Kamaljeet Singh (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

DOI: 10.1103/lxx5-43qg (publication)

Published in: Phys. Rev. D 112, 034032 (2025)

Impact factor: 5.3

- Magneto-Thomson and transverse Thomson effects in an interacting hadron gas in the presence of an external magnetic field

Kamaljeet Singh (Indian Inst. Tech., Indore), Kshitish Kumar Pradhan (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

DOI: 10.1103/crwt-166d (publication)

Published in: Phys. Rev. D 113, 034021 (2026)

Impact factor: 5.3

- Thermoelectric figure of merit and the deconfinement phase transition

Kamaljeet Singh (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

DOI:10.1016/j.physletb.2025.140060 (publication)

Phys. Lett. B 872, 140060 (2026)

Impact factor: 4.5

- Soret and Dufour effects in hot and dense QCD matter

Kamaljeet Singh (Indian Inst. Tech., Indore), Kangkan Goswami (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

DOI: 10.1103/q97t-xs31 (publication)

Published in: Phys. Rev. D 113, 014040 (2026)

Impact factor: 5.3

- Development of next-generation light-weight ternary Mg–Al–Li alloys for beampipe applications in particle accelerators

Kamaljeet Singh (Indian Inst. Tech., Indore), Kangkan Goswami (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore), Sumanta Samal (Indian Inst. Tech., Indore) [arXiv:2601.12836 [physics.acc-ph]] (Under Review)

- Thermodiffusive coupled-transport phenomena in dense quark matter

Kamaljeet Singh (Indian Inst. Tech., Indore), Kangkan Goswami (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore) [arXiv: 2606.04539 [hep-ph]] (Under Review)

CONFERENCE PROCEEDINGS

- Effect of time-varying electromagnetic field on Wiedemann-Franz law in a hot hadronic matter

Kamaljeet Singh (Indian Inst. Tech., Indore), Jayanta Dey (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore), Sabyasachi Ghosh (Indian Inst. Tech., Bhubaneswar)

Published in: DAE Symp.Nucl.Phys. 67, 891 (2024)

- Effect of temperature evolution on the thermal conductivity of magnetized quark-gluon plasma

Jayanta Dey (Indian Inst. Tech., Indore), **Kamaljeet Singh** (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

Published in: DAE Symp.Nucl.Phys. 67, 1073 (2024)

- Induced electromagnetic field in an evolving quark-gluon plasma due to thermoelectric effects

Kamaljeet Singh (Indian Inst. Tech., Indore), Jayanta Dey (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

Published in: DAE Symp. Nucl. Phys. 68, 999 (2025)

- Induced electric field due to thermoelectric effects in evolving quark-gluon plasma formed in heavy-ion collisions

Kamaljeet Singh (Indian Inst. Tech., Indore), Jayanta Dey (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

Published in: J. Subatomic Part. Cosmol. 4, 100112 (2025)

- Electrical conductivity of hadronic matter in a time-varying magnetic field

Jayanta Dey (Indian Inst. Tech., Indore), **Kamaljeet Singh** (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

Published in: J. Subatomic Part. Cosmol. 4, 100219 (2025)

- Induced electric field due to thermoelectric effects in an evolving quark-gluon plasma

Kamaljeet Singh (Indian Inst. Tech., Indore), Jayanta Dey (Indian Inst. Tech., Indore), Raghunath Sahoo (Indian Inst. Tech., Indore)

Springer Proc. Phys. 432, 835-839 (2026)

PRESENTATIONS

INTERNATIONAL CONFERENCES

- Poster Presentation at **ICPAQGP-2023 conference**, Puri, Odisha, India (07-10 Feb. 2023)
- Oral Presentation at **10TH ATHIC**, IISER Beharmpur, India (13-16 Jan. 2025)
- Poster Presentation at **Initial Stages 2025**, Taipei, Taiwan (07-12 Sep. 2025)
- Flash talk at **ALICE Physics week**, CERN, Geneva (03-07 Nov. 2025)

NATIONAL CONFERENCES

- Poster Presentation at **67th DAE Symposium on Nuclear Physics**, IIT Indore, India (09-13 Dec. 2023)
- Poster Presentation at **68th DAE Symposium on Nuclear Physics**, IIT Roorkee, India (07-11 Dec. 2024)
- Oral Presentation at **XXVI DAE-BRNS High Energy Physics Symposium 2024**, Banaras Hindu University, India (19-23 Dec. 2024)
- Oral Presentation at **HotQCD matter 2025 (Series 3)**, IIT Bhilai, India (04-05 Sep. 2025)

SCHOOLS and MEETINGS

- Attended **5th ALICE-STAR India school** on Quark-Gluon Plasma at Institute of Physics, Bhubaneswar, India in 2022
- Attended **ALICE-STAR collaboration meeting** at VECC Kolkata, India on september, 2022
- Attended **ALICE-STAR collaboration meeting** at Institute of Physics, Bhubaneswar, India, in June 2024
- Attended **Physics week** at CERN, Geneva in Nov. 2025
- Attended **ALICE week** at CERN, Geneva in Dec. 2025

TEACHING EXPERIENCES

- Teaching Assistant for BTech Students at IIT Indore, India.

Subject: Physics Lab, Duration: 4 semesters / 2 years (2022, 2023). I have explained the working theory and procedure of the experiments to the B.Tech. students in the physics laboratory. On the basis of the performance of the individual students, I have to provide the grades that further contribute to their final report

- Assistant Professor (Ad-hoc basis) - Guru Nanak National College, Panjab University, Chandigarh, India (2019-2021). I have taught the subjects - Quantum mechanics, Statistical mechanics, Thermodynamics and Nuclear and particle physics to undergraduate students

SKILLS and INTERESTS

- **Programming:** C, C++, Python
- **Event Generators:** PYTHIA (for proton-proton collision), A Multiphase Transport Model- AMPT (for heavy-ion collisions).
- **Reading** Enjoy reading books in English, Hindi, and Punjabi
- **Sports and Activities** Playing badminton, swimming and running
- **Nature Lover** Love being in natural surroundings and having reflective conversations with myself